MOBAN — Exam June 2025

Exercise 1: Linear Programming — Simplex Algorithm

Problem

```
max z = x1 + 3x2 − 4x3 + x4
s.t.   x1 + x2         ≤ 7
              x3        ≤ 9
       x1 − x2 + x4    ≤ 5
       x1, x2, x3, x4 ≥ 0
```

Step 1: Convert to Standard Form

Add slack variables s1, s2, s3 ≥ 0 to turn each inequality into an equality:

```
x1 + x2              + s1           = 7
        x3                 + s2      = 9
x1 − x2 + x4               + s3     = 5
```

Initial Basic Feasible Solution (BFS):

- Basic variables: s1 = 7, s2 = 9, s3 = 5
- Non-basic (= 0): x1 = 0, x2 = 0, x3 = 0, x4 = 0
- z = 0

Write the objective row as: **z − x1 − 3x2 + 4x3 − x4 = 0**

(Positive z-row coefficient = increasing that variable hurts z → don't enter. Negative = beneficial → enter.)

> **Note on x3:** Its z-row coefficient is already **+4** in the initial tableau → increasing x3 decreases z → x3 = 0 throughout all iterations.

Initial Tableau

BV	x1	x2	x3	x4	s1	s2	s3	RHS
s1	1	**1**	0	0	1	0	0	7
s2	0	0	1	0	0	1	0	9
s3	1	−1	0	1	0	0	1	5

BV	x1	x2	x3	x4	s1	s2	s3	RHS
z	−1	**−3**	4	−1	0	0	0	0

Iteration 1

Entering variable: x2 — most negative z-row coefficient (−3)

Min-ratio test (only rows with positive coefficient in the x2 column):

Row	RHS	x2 coeff	Ratio
s1	7	1	7/1 = **7** ← min
s2	9	0	— (skip)
s3	5	−1	— (negative, skip)

Leaving variable: s1 | **Pivot element:** row 1, column x2, value = 1

Row operations:

```
New x2-row  = Row1 ÷ 1              (no change, pivot = 1)
New s2-row  = Row2 − 0 × Row1      (no change, x2 coeff = 0)
New s3-row  = Row3 − (−1) × Row1   = Row3 + Row1
New z-row   = z-row − (−3) × Row1  = z-row + 3 × Row1
```

Tableau after Iteration 1:

BV	x1	x2	x3	x4	s1	s2	s3	RHS
x2	1	1	0	0	1	0	0	7
s2	0	0	1	0	0	1	0	9
s3	2	0	0	**1**	1	0	1	12
z	2	0	4	**−1**	3	0	0	21

Current BFS: x2 = 7, s2 = 9, s3 = 12, **z = 21**

Iteration 2

Entering variable: x4 — only remaining negative z-row coefficient (−1)

Min-ratio test:

Row	RHS	x4 coeff	Ratio
x2	7	0	— (skip)

Row	RHS	x4 coeff	Ratio
s2	9	0	— (skip)
s3	12	1	12/1 = **12** ← only option

Leaving variable: s3 | **Pivot element:** row 3, column x4, value = 1

Row operations:

```
New x4-row  = Row3 ÷ 1              (no change, pivot = 1)
New x2-row  = Row1 − 0 × Row3      (no change, x4 coeff = 0)
New s2-row  = Row2 − 0 × Row3      (no change, x4 coeff = 0)
New z-row   = z-row − (−1) × Row3  = z-row + Row3
```

Tableau after Iteration 2 — OPTIMAL:

BV	x1	x2	x3	x4	s1	s2	s3	RHS
x2	1	1	0	0	1	0	0	7
s2	0	0	1	0	0	1	0	9
x4	2	0	0	1	1	0	1	12
z	**4**	0	**4**	0	**4**	0	**1**	**33**

All z-row coefficients ≥ 0 → **no further improvement possible** → **OPTIMAL.**

Optimal Solution

Variable	Value
x1	0
x2	**7**
x3	0
x4	**12**
z*	**33**

Verification

```
z = x1 + 3x2 − 4x3 + x4
  = 0 + 3(7) − 4(0) + 12
  = 21 + 12
  = 33 ✓
```

```
Constraint 1:  x1 + x2       = 0 + 7 = 7  ≤ 7  ✓  (tight)
Constraint 2:  x3             = 0         ≤ 9  ✓
Constraint 3:  x1 − x2 + x4  = 0 − 7 + 12 = 5  ≤ 5  ✓  (tight)
```

Intuition Behind the Solution

Variable	Why this value?
x2 = 7	Coefficient +3 in objective (highest) → push to max. Constraint 1 limits it to 7.
x4 = 12	Coefficient +1 in objective → push to max. With x2 = 7, constraint 3 allows x4 up to 0 − 7 + x4 ≤ 5, so x4 ≤ 12.
x1 = 0	Positive objective coefficient (+1), but entering x1 would force x2 or x4 to decrease — net effect negative in the final tableau (reduced cost = −4).
x3 = 0	Coefficient −4 in objective → increasing x3 always hurts z; never enters the basis.

Summary: Simplex Pivots

Iteration	Entering	Leaving	Pivot row	z value
Initial	—	—	—	0
1	x2	s1	Row 1	21
2	x4	s3	Row 3	33
Optimal	—	—	—	**33**

Exercise 2: EPON — IPACT Limited DBA

System Parameters

Parameter	Value
N (ONUs)	8
Distance OLT–ONU (d)	10 km
W_max	10,000 bytes
Guard time (T_g)	2 μs
Buffer size	1 MB
Max Ethernet frame	1,518 bytes
REPORT / GATE size	64 bytes
Upstream line rate (R)	2 Gbps

Parameter	Value
Traffic	Uniform load, memoryless (Poisson)

Pre-computed constants:

```
v = 300,000 km/s / 1.5 = 200,000 km/s   (speed in optical fiber)
RTT = 2 × 10 / 200,000 = 100 μs
T_Wmax = 10,000 × 8 / 2×10⁹ = 40 μs
T_ctrl = 64 × 8 / 2×10⁹ = 0.256 μs   (REPORT or GATE transmission time)
```

(a) Minimal Cycle Length

The minimum possible cycle is RTT-limited: the OLT cannot send the next GATE until it has received the REPORT from the current cycle, which requires one full round trip.

```
T_cycle_min = RTT + T_report + T_gate
            = 100 + 0.256 + 0.256
            ≈ 100.5 μs
```

(b) Average Cycle Length at 95 % Load

Using the load-balance equation for IPACT Limited (Gated-style derivation):

- Cycle structure: `T = N × (X + T_report + T_g)` where X = data time per ONU
- Load definition: `L = N × X / T` → `N × X = L × T`
- Substituting: `T = L×T + N×(T_report + T_g)` → `T(1−L) = N×(T_report + T_g)`

```
T_cycle = N × (T_report + T_g) / (1 − L)
        = 8 × (0.256 + 2) / (1 − 0.95)
        = 8 × 2.256 / 0.05
        = 18.048 / 0.05
        ≈ 361 μs
```

Check: 361 μs > T_cycle_min = 100.5 μs ✓

(c) Average PON Packet Delay at 95 % Load

The average PON packet delay = queuing/waiting time at ONU + propagation from ONU to OLT.

Average waiting time in an IPACT polling system (packet arrives uniformly within the cycle):

```
W_avg = 1.5 × T_cycle = 1.5 × 361 = 541.5 µs
```

One-way propagation delay:

```
T_prop = d / v = 10 / 200,000 = 50 µs
```

T_W?

```
T_Wmax = 10kB/2Gbps = 40us

or

0.95 = 8*T_P/8*(T_P + T_R + T_G)
=> T_P = 0.95(T_R (2us) + T_G (0.25us) )/(1-0.95) = 42,75
```

Total average PON packet delay:

```
D_avg = W_avg + T_prop + T_Wmax/2 = 541.5 + 50 + 20 ≈ 612 µs
```

(d) Highest Achievable Load for W_max = 5,000 bytes

When W_max is capped at 5,000 bytes, every ONU can transmit at most W_max bytes per cycle. The system reaches saturation when all ONUs always fill their window:

```
T_Wmax_new = 5,000 × 8 / 2×10⁹ = 20 µs

L_max = T_Wmax/(T_Wmax + T_R + T_G)
```

> **Reducing W_max from 20,000 → 5,000 bytes lowers the maximum achievable load from ~97 % to ~89.9 %.** A smaller window means the guard-time and control overhead represent a larger fraction of each slot, reducing efficiency.

Exercise 3: Wi-Fi 7 — Maximum Physical Bitrate

System Parameters

Parameter	Value
Modulation	4096-QAM → 12 bits/subcarrier
Coding rate	5/6
OFDM symbol time	12.8 μs
Guard Interval options	0.8 μs, 1.6 μs, 3.2 μs
OFDMA	4 × 980-tone RU (320 MHz band)
Spatial streams	8

(a) Maximum Physical Bitrate

For maximum bitrate, choose the **shortest Guard Interval = 0.8 μs**.

Total OFDM symbol duration:

```
T_symbol = 12.8 + 0.8 = 13.6 μs
```

Total data subcarriers:

```
N_data = 4 RUs × 980 tones = 3,920 subcarriers
```

Bits per OFDM symbol (across all streams):

```
bits/symbol = N_data × bits/subcarrier × coding_rate × N_SS
            = 3,920 × 12 × (5/6) × 8
            = 3,920 × 10 × 8
            = 313,600 bits/symbol
```

Maximum physical bitrate:

```
R_max = bits/symbol / T_symbol
      = 313,600 / (13.6 × 10⁻⁶)
      ≈ 23.06 Gbps
```

Guard Interval	T_symbol	Bitrate
0.8 μs	13.6 μs	**23.06 Gbps** ← maximum
1.6 μs	14.4 μs	21.78 Gbps
3.2 μs	16.0 μs	19.60 Gbps

Exercise 4: Wi-Fi Network Design

(a) Application Traffic Characteristics

Video stream:

- Application data per frame: 7,300 bytes + 12 bytes RTP header = 7,312 bytes
- UDP payload per IP packet: 1,500 − 20 (IP) − 8 (UDP) = 1,472 bytes
- IP packets per video frame: $\lceil 7{,}312 / 1{,}472 \rceil$ = **5 IP packets**
  - 4 × full: 20 + 8 + 1,472 = **1,500 bytes**
  - 1 × last: 20 + 8 + (7,312 − 4 × 1,472) = 20 + 8 + 1,424 = **1,452 bytes**

Monitoring:

- IP packet: 72 (data) + 8 (UDP) + 20 (IP) = **100 bytes** → fits in one IP packet

Application	IP Packet Size	DL (AP→STA) / UL (STA→AP)	Traffic behavior
Video	1,500 bytes (max), 5 packets/frame	**DL** (AP → STA)	CBR, periodic at 50 fps
Monitoring	100 bytes	**UL** (STA → AP)	Periodic, 1 msg / 100 ms per STA

(b) T_Data Expression

For one PPDU carrying N aggregated packets of size P bytes at PHY rate R, using HT Greenfield format.

HT Greenfield preamble structure:

```
| HT-GF-STF | HT-LTF 1 | HT-SIG | HT-LTF 2 … N_DLTF |  DATA  |
|   8 µs    |   8 µs   |  8 µs  |  (N_DLTF−1) × 4 µs |        |
```

The number of data LTFs N_DLTF = smallest value in {1, 2, 4} ≥ N_SS:

N_SS	N_DLTF	Extra LTFs after SIG	Extra time
1	1	0	0 µs
2	2	1	4 µs
3 or 4	4	3	12 µs

This exercise specifies "No spatial multiplexing" → N_SS = 1 → N_DLTF = 1 → no extra HT-LTFs after HT-SIG.

Total preamble = 8 + 8 + 8 + 0 = **24 µs**

```
T_Data = 24 μs + (N × P × 8) / R
```

where P includes the full MPDU payload (IP + LLC + MAC header + FCS).

(c) Network Design

Step 1 — Channel Band

Choose 5 GHz.

Justification: 5 GHz has less interference than 2.4 GHz (fewer overlapping networks, less congestion from Bluetooth/microwave). Minimum MCS 3 (16-QAM, 1/2 coding) is guaranteed for all STAs.

Step 2 — Bandwidth

Choose 40 MHz.

From the MCS table, 5 GHz, MCS 3, Short GI, 40 MHz: R = **60 Mbps**.

This gives sufficient capacity with room for multiple streams and measurement STAs.

Step 3 — Timing verification (using T_Data formula)

Video DL — A-MPDU with 5 MPDUs per video frame:

```
P_MPDU = LLC(8) + IP(1500) + MAC_header(24) + FCS(4) = 1,536 bytes
Total = 5 × 1,536 = 7,680 bytes

T_Data = 24 μs + 7,680 × 8 / (60 × 10⁶)
       = 24 + 1,024
       = 1,048 μs ≈ 1.05 ms   ≪ 200 ms delay budget ✓
```

Monitoring UL — one IP packet per poll, PCF polling:

```
P_MPDU = LLC(8) + IP(100) + MAC_header(24) + FCS(4) = 136 bytes

T_Data = 24 + 136 × 8 / (60 × 10⁶) = 24 + 18.1 = 42.1 μs

Per STA: T_POLL + SIFS + T_Data + SIFS = 40 + 16 + 42.1 + 16 = 114.1 μs
200 STAs: 200 × 114.1 = 22,820 μs = 22.8 ms   < 100 ms delay budget ✓
```

Design Summary

Config	Choice	Justification
Channel	**5 GHz**	Less interference; MCS 3 guaranteed; suitable for high-rate short-range links
Bandwidth	**40 MHz**	60 Mbps at MCS 3; handles video + 200 monitoring STAs within delay budget
MAC mechanism	**DCF (CSMA/CA + RTS/CTS)** for video DL; **PCF** for monitoring UL	PCF polling ensures bounded UL delay for 200 STAs; RTS/CTS avoids hidden terminal for large video frames
Aggregation	**A-MPDU, 5 MPDUs** per video PPDU	Amortises 24 μs preamble over 5 packets; No ACK policy eliminates BlockACK overhead

---

(d) Saturation Check

Medium time consumed per second:

```
Video DL (with DIFS + backoff):
  DIFS = 16 + 2×9 = 34 μs; avg backoff = (15/2)×9 = 67.5 μs
  Per PPDU: 34 + 67.5 + 1,048 = 1,149.5 μs
  50 frames/s: 50 × 1,149.5 = 57,475 μs = 57.5 ms/s

Monitoring UL (PCF, no contention overhead):
  200 STAs polled every 100 ms → 10 rounds/s
  Per round: 200 × 114.1 μs = 22,820 μs = 22.8 ms
  Per second: 10 × 22.8 = 228 ms/s
```

Total used: 57.5 + 228 = **285.5 ms/s**

Remaining capacity: 1,000 − 285.5 = **714.5 ms/s** (≈ 71.5 % free)

The network is NOT saturated.

Additional capacity available:

- Each extra video stream uses ≈ 57.5 ms/s → room for **≈ 12 more streams**
- Each extra monitoring STA uses ≈ 10 × 114.1 μs = 1.14 ms/s → room for **hundreds more STAs**

Exercise 5: Cellular Networks — GSM & Spectrum Regulation

(a) Simultaneous Full-Rate Calls in One GSM Cell

Total physical channels:

```
16 frequencies × 8 time slots/frequency = 128 physical channels
```

Channels reserved for broadcast and common control:

```
5 time slots reserved for BCCH, FCCH, SCH, CCCH (PAGCH)
```

Channels needed for SDCCH (call setup signaling — 50 simultaneous setups required):

Each SDCCH/8 sub-channel uses 1 physical time slot and supports 8 parallel setups.

```
Number of SDCCH/8 slots = ⌈50/8⌉ = 7 slots  (7 × 8 = 56 ≥ 50)
```

TCH channels available for active voice calls:

```
TCH = 128 − 5 (broadcast/control) − 7 (SDCCH)
    = 116 simultaneous full-rate calls
```

(b) Cellular vs. Wi-Fi — Spectrum Regulation & Radio Resource Management

	Cellular (4G/5G)	Wi-Fi
Spectrum regulation	Licensed, exclusive spectrum — operators purchase frequency bands at government auctions. Each operator owns their band; no one else can transmit in it.	Unlicensed, shared spectrum (ISM/UNII bands). Anyone can transmit without a licence, subject to power limits and etiquette rules (e.g., ETSI EN 300 328).
Radio resource management	Centralized: the base station (eNodeB/gNB) schedules every uplink and downlink resource block per slot. Guarantees QoS per user; inter-cell interference is managed by the network via frequency planning, power control, and coordination (X2 interface).	Distributed: CSMA/CA contention-based — each device decides independently when to transmit. No QoS guarantee (EDCA gives statistical priority only). Interference from neighboring APs/STAs is uncontrolled; adjacent deployments cannot coordinate.

Consequences:

- Cellular operators can guarantee service quality and plan coverage/capacity, but spectrum licensing is expensive → cost passed on to users via subscriptions.
- Wi-Fi is free to deploy but QoS degrades in dense environments where many devices/APs compete for the same unlicensed channel.

(c) "Cellular and Wi-Fi standards are converging"

Several dimensions of convergence:

1. **Physical layer convergence:** Both now use OFDM/OFDMA modulation, MIMO, and beamforming. Wi-Fi 6 (802.11ax) borrowed OFDMA directly from LTE/5G; 5G NR adopted flexible numerology inspired by Wi-Fi's OFDM design.

2. **Feature convergence:** Cellular brought carrier aggregation and QoS bearers; Wi-Fi responded with EDCA (QoS), Block ACK, MU-MIMO (802.11ac/ax). 5G NR-U (New Radio Unlicensed) runs 5G protocols in Wi-Fi's unlicensed 5 GHz / 6 GHz bands.

3. **Spectrum convergence:** 5G is entering unlicensed bands (Wi-Fi territory), while private LTE/5G networks use spectrum that was once reserved for cellular operators only. The 6 GHz band is now shared between Wi-Fi 6E and private 5G deployments.

4. **Architecture convergence:** 3GPP defines tight WLAN–cellular interworking (EPC-integrated Wi-Fi offload, Passpoint/Hotspot 2.0, VoWiFi). Operators seamlessly hand traffic between cellular and Wi-Fi without the user noticing.

5. **Use case convergence:** Wi-Fi is used for voice calls (VoWiFi), video conferencing, and IoT — use cases once considered exclusively cellular. Cellular (NB-IoT, LTE-M) now targets low-power sensor deployments that were once only served by Zigbee/Wi-Fi.

In short: the boundary between "cellular" and "Wi-Fi" is blurring at the PHY, MAC, spectrum, and architecture levels.

Exercise 6: AODV in a Wireless Ad Hoc Network

Network Topology

```
        1 ── 2 ── 3
        |    |    |
   4 ── 5 ── 6 ── 7 ── 8
        |    |    |
        9 ── 10── 11
```

Node 6 is the server. All 10 other nodes want to connect to Node 6.

Hop distances from Node 6:

Source node	Shortest path to 6	Hops (h)
1	6→5→1	2
2	6→2	1

Source node	Shortest path to 6	Hops (h)
3	6 → 7 → 3	2
4	6 → 5 → 4	2
5	6 → 5	1
7	6 → 7	1
8	6 → 7 → 8	2
9	6 → 5 → 9	2
10	6 → 10	1
11	6 → 7 → 11	2

(a) Worst-Case Total Routing Control Overhead

In AODV, for each route discovery:

- **RREQs:** (N – 1) = 10 broadcasts (every node except destination rebroadcasts once)
- **RREPs:** h unicast forwards back to the source (one per hop on the return path)

With **10 sources** connecting simultaneously (worst case = all at once, no caching):

```
Total RREQs = 10 sources × 10 RREQs each = 100 RREQ transmissions

Total RREPs = Σ h_i = 2+1+2+2+1+1+2+2+1+2 = 16 RREP transmissions

Total control overhead = 100 + 16 = 116 control packets
```

(b) Why Control Overhead Can Be Lower in Reality

1. **Route caching / intermediate RREP:** If a node already has a valid, unexpired routing table entry for Node 6 (learned from a previous RREQ flood), it can reply with an RREP immediately without forwarding the RREQ further. This cuts the RREQ flood short.

2. **Simultaneous setup reuse:** When 10 RREQs flood the network simultaneously, intermediate nodes that forward RREQ_A (from node 1) also record a reverse route back to node 1. Node 6 can then piggyback replies, and intermediate nodes gain routing knowledge for free.

3. **AODV route entry lifetime:** After the first connection is established, routing table entries remain valid for a TTL period. Subsequent connections from the same source reuse the cached route without re-flooding.

4. **Expanding ring search (TTL limiting):** If the destination is nearby, the RREQ is stopped before reaching all N nodes, dramatically reducing RREQs for short paths.

(c) Method Wi-Fi Uses to Avoid RREQ Collision

Wi-Fi uses **random jitter** before rebroadcasting: when a node receives a RREQ and decides to forward it, it waits a **random backoff delay** before actually transmitting. This desynchronises the rebroadcasts from multiple neighbours, reducing the probability that two nodes transmit simultaneously and collide.

This is the standard IEEE 802.11 MAC collision-avoidance mechanism applied to broadcast RREQ propagation — a mitigation for the **broadcast storm problem**.

(d) Effect of Using Average Connection Speed as Route Metric

Standard AODV selects the route whose RREQ **arrives first** (minimum delay metric). Switching to **average connection speed** changes the procedure as follows:

1. **RREQ carries accumulated metric:** Each intermediate node appends the speed of the incoming link to the RREQ. The metric propagated is the minimum (bottleneck) link speed along the path so far.

2. **Destination cannot reply immediately:** Node 6 must wait to receive RREQs from multiple paths before it knows which path has the best metric. It sets a timer and waits for additional RREQs before sending the RREP.

3. **Multiple RREPs may be sent:** As better-metric RREQs arrive, Node 6 may send a new RREP superseding an earlier one. Intermediate nodes update their routing tables if the new RREP carries a better metric.

4. **Higher setup latency:** The connection setup is delayed because the destination waits for the best-metric RREQ rather than responding to the first one.

5. **Better path quality:** The selected route maximises bottleneck link speed, improving throughput at the cost of longer discovery time and potentially more control traffic.